

Amylase and pH: Continuous-Sampling Investigation

Today I can... ☐ Order the preparation, timing and sampling steps, and justify one evidence decision ☐

Identify the first observed iodine endpoint and record the time with its sampling limit ☐ Use matched conditions and raw observations to make a supported comparison

Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R _____



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LAUNCH SCIENCE | last lesson |

Amylase sampling investigation

Knowledge organiser

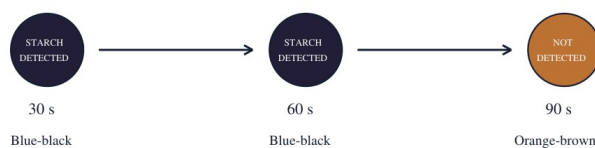
Learning objective

I can use continuous-sampling evidence to investigate how pH affects amylase activity.

Success criteria

1. Order the preparation, timing and sampling steps, and justify one evidence decision.
2. Identify the first observed iodine endpoint and record the time with its sampling limit.
3. Use matched conditions and raw observations to make a supported comparison.

Supplied observations: pH 7, one illustrative trial, 30 s intervals.



First observed endpoint = 90 s. The endpoint lies after 60 s and at or before 90 s.

Read a tested sample

Continuous sampling means testing separate small samples at fixed intervals while the reaction continues. Blue-black means starch detected; orange-brown remaining means starch not detected in that drop.

Keep the comparison valid

Change pH: measure endpoint time / s. Match temperature, enzyme and starch amounts, volumes and the endpoint rule. Prepare labelled iodine wells before mixing; keep iodine out of the main reaction.

State the timing limit

The 30 s interval gives a window, not an exact midpoint. If the 60 s sample were missing, 30 s blue-black and 90 s orange-brown would give a 60 s window. Preserve a missed sample or uncertain colour.

Use a bounded conclusion

First-series endpoint times: pH 4 = 240 s; 5 = 180 s; 6 = 120 s; 7 = 90 s; 8 = 150 s. pH 7 is fastest in this illustrative series. One trial per pH does not prove an exact universal optimum.

Made by Matt | Use the reference to explain, calculate and evaluate.

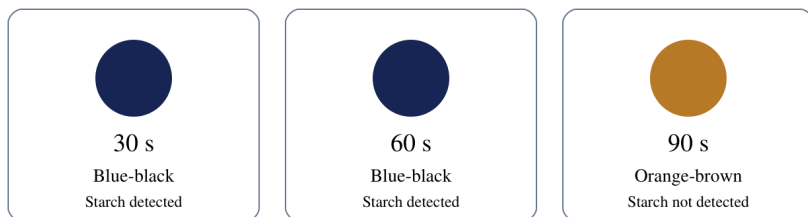
Practical reference data and models

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Enzyme: biological catalyst; speeds a reaction without being used up. Substrate: substance it acts on. pH: a measure of acidity or alkalinity. Amylase acts on starch, producing smaller sugars.

Iodine key: blue-black = starch detected. Orange-brown remains = starch not detected. check separate sampled drops.

Endpoint = first sampled drop leaving iodine orange-brown; record its time / s.



Method card	Action
1	Prepare and check: approved setup, labelled wells, timer and roles ready.
2	Start together: mix and start timing as the approved method states.
3	Sample at fixed times: separate samples into prepared iodine wells.
4	Keep raw observations: record colour and time honestly as sampling continues.

Supplied illustrative observations — not your practical results. Single runs; equal starting amounts and the same iodine endpoint. Samples every 30 s. pH 7: blue-black at 30/60 s, orange-brown at 90 s. Record 90 s.

pH	First endpoint / s
4	240
5	180
6	120
7	90
8	150

Valid comparison: change pH; match temperature, enzyme and starch concentrations/volumes, total volume, starting and sampling conditions. A separate no-active-amylase control should remain starch-positive. This is different from a factor kept constant.

A separate pH 7 repeat gave 120 s: keep both 90 s and 120 s. Another run had no endpoint by 300 s: record that observation, not zero. A paper investigation uses supplied observations; it does not show that you completed reagent handling.